TRIGONOMETRIC IDENTITIES

Pythagorean Identities

$$\sin^2(A)+\cos^2(A)=1 \qquad 1+\tan^2(A)=\sec^2(A) \qquad 1+\cot^2(A)=\csc^2(A)$$

Quotient Identities

$$\csc(A)=\frac{1}{\sin(A)} \qquad \sec(A)=\frac{1}{\cos(A)} \qquad \tan(A)=\frac{\sin(A)}{\cos(A)} \qquad \cot(A)=\frac{1}{\tan(A)}=\frac{\cos(A)}{\sin(A)}$$

Co-function Identities (Complementary angles)

$$\sin\left(\frac{\pi}{2}-A\right)=\cos(A) \qquad \sec\left(\frac{\pi}{2}-A\right)=\csc(A) \qquad \tan\left(\frac{\pi}{2}-A\right)=\cot(A)$$

Negative Angle Identities

$$\sin(-A)=-\sin(A) \qquad \cos(-A)=\cos(A) \qquad \tan(-A)=-\tan(A)$$
$$\csc(-A)=-\csc(A) \qquad \sec(-A)=\sec(A) \qquad \cot(-A)=-\cot(A)$$

Double Angle Identities

$$\sin(2A)=2\sin(A)\cos(A) \qquad \cos(2A)=\cos^2(A)-\sin^2(A)=1-2\sin^2(A)=2\cos^2(A)-1 \qquad \tan(2A)=\frac{2\tan(A)}{1-\tan^2(A)}$$

Half Angle Identities

$$\sin\left(\frac{A}{2}\right)=\pm\sqrt{\frac{1-\cos(A)}{2}} \qquad \cos\left(\frac{A}{2}\right)=\pm\sqrt{\frac{1+\cos(A)}{2}} \qquad \tan\left(\frac{A}{2}\right)=\pm\sqrt{\frac{1-\cos(A)}{1+\cos(A)}}$$

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Angle Addition and Subtraction Identities

$$\sin(A+B)=\sin(A)\cos(B)+\cos(A)\sin(B) \qquad \sin(A-B)=\sin(A)\cos(B)-\cos(A)\sin(B)$$

$$\cos(A+B)=\cos(A)\cos(B)-\sin(A)\sin(B) \qquad \cos(A-B)=\cos(A)\cos(B)+\sin(A)\sin(B)$$

$$\tan(A+B)=\frac{\tan(A)+\tan(B)}{1-\tan(A)\tan(B)} \qquad \tan(A-B)=\frac{\tan(A)-\tan(B)}{1+\tan(A)\tan(B)}$$

Product Identities

$$\sin(A)\cos(B)=\frac{1}{2}\left(\sin(A+B)+\sin(A-B)\right) \qquad \cos(\mathrm{A})\sin(\mathrm{B})=\frac{1}{2}\left(\sin(A+B)-\sin(A-B)\right)$$

$$\sin(A)\sin(B)=\frac{1}{2}\left(\cos(A-B)-\cos(A+B)\right) \qquad \cos(\mathrm{A})\cos(\mathrm{B})=\frac{1}{2}\left(\cos(A+B)+\cos(A-B)\right)$$

Sum and Difference Identities

$$\sin(A)+\sin(B)=2\sin\left(\frac{A+B}{2}\right)\cos\left(\frac{A-B}{2}\right)$$

$$\sin(A)-\sin(B)=2\cos\left(\frac{A+B}{2}\right)\sin\left(\frac{A-B}{2}\right)$$

$$\cos(A)+\cos(B)=2\cos\left(\frac{A+B}{2}\right)\cos\left(\frac{A-B}{2}\right)$$

$$\cos(A)-\cos(B)=-2\sin\left(\frac{A+B}{2}\right)\sin\left(\frac{A-B}{2}\right)$$

Reduction Identities

$$A\sin(\theta)\pm B\cos(\theta)=\sqrt{A^2+B^2}\,\sin(\theta\pm\alpha)$$

where α is determined from

$$\sin(\alpha)=\frac{B}{\sqrt{A^2+B^2}} \qquad \cos(\alpha)=\frac{A}{\sqrt{A^2+B^2}}$$

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